Mid-term Solution

EE241 , Spring 2019

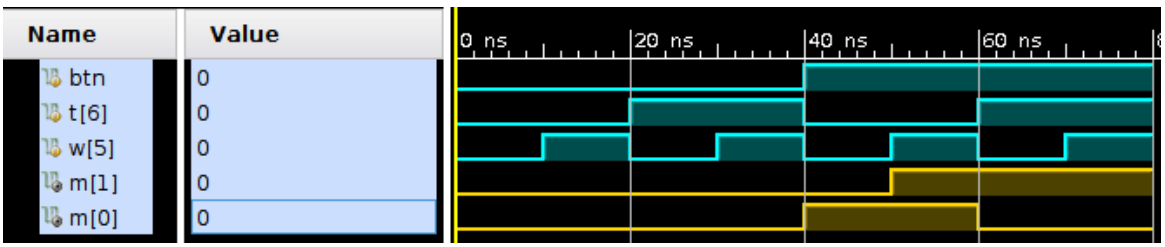
Q1 (30 pts.)

btn	t[6]	w[5]	m[1]	m[0]
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	1
1	0	1	1	1
1	1	0	1	0
1	1	1	1	0

```
‘timescale 1ns / 1ps

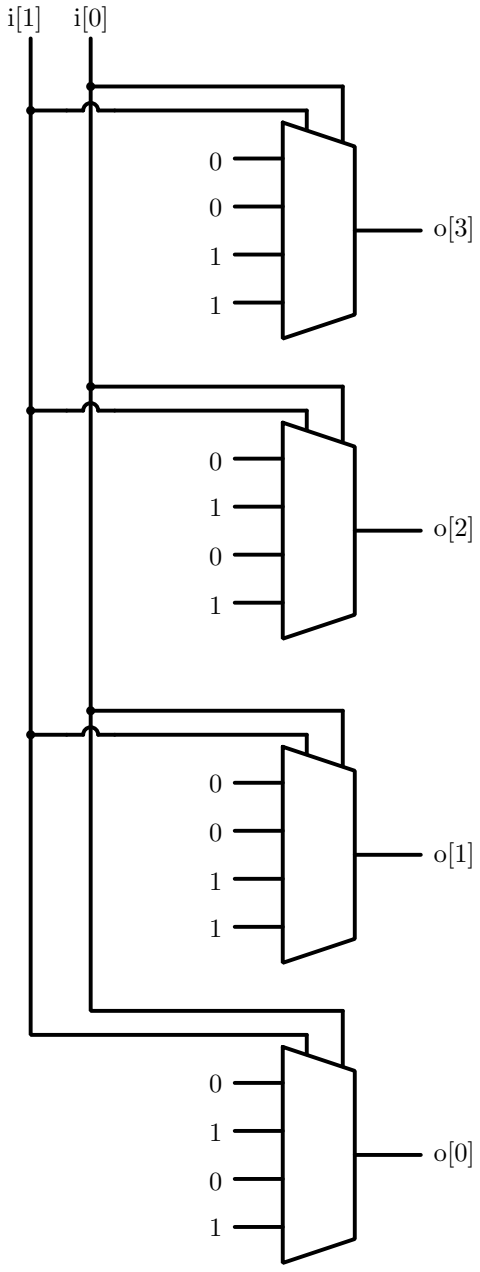
module q1(btn,t,w,m);
input btn;
input [6:0] t;
input [5:0] w;
output [1:0] m;
assign m[1]=btn&(t[6]|w[5]);
assign m[0]=btn&~t[6];
endmodule

module q1_tb;
reg btn;
reg [6:0] t;
reg [5:0] w;
wire [1:0] m;
q1 uut(btn,t,w,m);
initial begin
{btn,t,w}={1’b0,7’d0,6’d0}; #10;
{btn,t[6],w[5]}=3’d1; #10;
{btn,t[6],w[5]}=3’d2; #10;
{btn,t[6],w[5]}=3’d3; #10;
{btn,t[6],w[5]}=3’d4; #10;
{btn,t[6],w[5]}=3’d5; #10;
{btn,t[6],w[5]}=3’d6; #10;
{btn,t[6],w[5]}=3’d7; #10; $stop;
end
endmodule
```



Q2 (30 pts.)

i[1]	**i[0]**	**o[3]**	**o[2]**	**o[1]**	**o[0]**
0	0	0	0	0	0
0	1	0	1	0	1
1	0	1	0	1	0
1	1	1	1	1	1



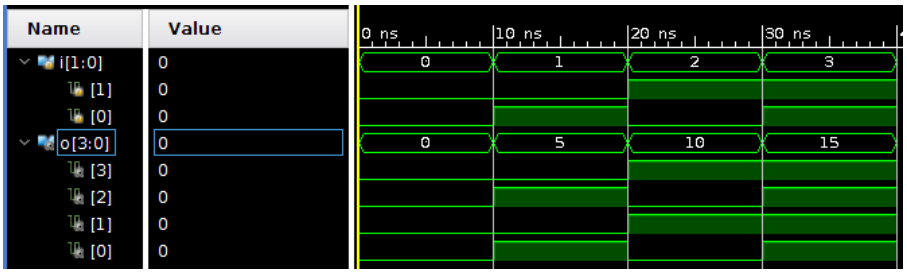
Bonus

```
‘timescale 1ns / 1ps

module q2(i,o);
input [1:0] i;
output [3:0] o;
assign o={i,i};
endmodule

module q2_tb;
reg [1:0] i;
wire [3:0] o;
q2 uut(i,o);

initial begin
i=2’b00; #10;
i=2’b01; #10;
i=2’b10; #10;
i=2’b11; #10; $stop;
end
endmodule
```



Q3 (40 pts.)

```
‘timescale 1ns / 1ps

module q3(num,shift,out);
input [7:0] num;
input [3:0] shift;
output [7:0] out;
assign out=(num+shift)*(1-num[0]) + num[0]*(~num+1);
endmodule

module q3_tb;
reg [7:0] num;
reg [3:0] shift;
wire [7:0] out;
q3 uut(num,shift,out);

initial begin
num=8’d98; // ascii ’b’
shift=4’d3; #10;
num=8’d99; // ascii ’c’
shift=4’d3; #10; $stop;
end
endmodule
```

